

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0294

Basilone 138 kV
October 31, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0294. The Interconnection Customer proposes a 58.3 MW solar photovoltaic facility interconnecting at the Basilone 138 kV substation in the West Virginia area. The request was received on November 29, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 216 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,502,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Blue / Black Rock 138 KV Line is overloaded to 165.8% of its 192 MVA emergency rating for the loss of the Faraday / Basilone Substation 138 KV. The project contributes 5.3% DFAX (3.1 MW) to this overload.

The Nashua / Black Rock 138 KV is overloaded to 163.3% of its 204 MVA emergency rating for the loss of the Faraday / Basilone 138 KV. The project contributes 5.3% DFAX (3.1 MW) to this overload.

Affinity / Batesville 138 KV line: 158.3% loading (150 MVA) following the outage of the Kean Mtn And Grassy Crk / Barr Lake 345 KV line; DFAX = 18.2%.

The Taloga / Parke 138 KV is overloaded to 136.9% of its 270 MVA emergency rating for the loss of the New Hope Ap / Barr Lake 345 KV. The project contributes 10.2% DFAX (6.0 MW) to this overload.

Coal Creek / Sandy Ridge 138 KV: 132.5% loading (236 MVA) following the outage of the Clinch Riv / Kean Mtn And Grassy Crk Substation 345 KV; DFAX = 13.5%.

For the loss of Phlan / Bradley 138 KV, the American Falls / Falconer Substation 138 KV line loads to 131.0% of its 191 MVA rating. Project DFAX: 5.4%.

The Sandy Ridge / N Reading 138 KV is overloaded to 126.4% of its 242 MVA emergency rating for the loss of the Clinch Riv / Kean Mtn And Grassy Crk 345 KV. The project contributes 13.5% DFAX (7.9 MW) to this overload.

The Reata / American Falls Substation 138 KV line is overloaded to 126.0% of its 306 MVA emergency rating for the loss of the Kean Mtn And Grassy Crk / Barr Lake 345 KV L. The project contributes 8.1% DFAX (4.7 MW) to this overload.

Reata / Glenoveer 138 KV: 125.8% loading (308 MVA) following the outage of the Kean Mtn And Grassy Creek / Barr Lake 345 KV; DFAX = 8.1%.

The loco / North Reading 138 KV is overloaded to 125.6% of its 243 MVA emergency rating for the loss of the Kean Mtn And Grassy Crk / Clinch River 345 KV. The project contributes 13.5% DFAX (7.9 MW) to this overload.

Greens Bayou 69 / Glenoveer 138 KV: 125.3% loading (312 MVA) following the outage of the Kean Mtn And Grassy Crk / Barr Lake Substation 345 KV L; DFAX = 8.1%.

For the loss of Clinch Riv / Kean Mtn And Grassy Crk 345 KV, the Kean Mtn And Grassy Creek 138-1 kV Auto loads to 123.3% of its 471 MVA rating. Project DFAX: 24.5%.

Kean Mtn And Grassy Crk Substation 345/1 kV XFMR: 123.3% loading (471 MVA) following the outage of the Kean Mtn And Grassy Creek / Clinch Riv 345 KV; DFAX = 24.5%.

For the loss of Chatom / Dayton 345 KV, the Dayton 345 kV/138 kV Autotransformer loads to 117.7% of its 479 MVA rating. Project DFAX: 5.1%.

For the loss of Kean Mtn And Grassy Crk / Barr Lake 345 KV, the Winnebago Wind Power Project / Clinch River 138 KV loads to 116.6% of its 246 MVA rating. Project DFAX: 6.0%.

Poplar Gap / Sandy Hook 138 KV L: 115.9% loading (226 MVA) following the outage of the Barr Lake / Kean Mtn And Grassy Crk 345 KV line; DFAX = 18.2%.

The Wesmer / Itmann 138 KV is overloaded to 115.3% of its 234 MVA emergency rating for the loss of the Barr Lake / Kean Mtn And Grassy Crk 345 KV. The project contributes 18.2% DFAX (10.6 MW) to this overload.

Baileysville / Itmann 138 KV: 115.1% loading (236 MVA) following the outage of the Kean Mtn And Grassy Crk / Barr Lake 345 KV Line; DFAX = 18.2%.

No Troy / Lost Nation 138 KV: 115.0% loading (240 MVA) following the outage of the Barr Lake / Kean Mtn And Grassy Crk 345 KV; DFAX = 6.1%.

The Sandy Hook / Wesmer 138 KV is overloaded to 114.9% of its 230 MVA emergency rating for the loss of the Barr Lake / Kean Mtn And Grassy Crk 345 KV. The project contributes 18.2% DFAX (10.6 MW) to this overload.

For the loss of Kean Mtn And Grassy Crk / Clinch Riv Substation 345 KV, the Ioco Substation / Kean Mtn And Grassy Crk Substation 138 KV loads to 114.6% of its 256 MVA rating. Project DFAX: 13.5%.

For the loss of Faraday / Basilone 138 KV, the Claypool Hill / Coal Crk 138 KV L loads to 114.2% of its 328 MVA rating. Project DFAX: 63.6%.

The Lynn / Lost Nation 138 KV is overloaded to 114.1% of its 239 MVA emergency rating for the loss of the Kean Mtn And Grassy Creek / Barr Lake 345 KV. The project contributes 6.1% DFAX (3.6 MW) to this overload.

Winnebago Wind Power Project / Lebanon Substation 138 KV: 114.0% loading (242 MVA) following the outage of the Kean Mtn And Grassy Creek / Barr Lake 345 KV line; DFAX = 6.0%.

Rogue / Lynn 138 KV: 113.9% loading (235 MVA) following the outage of the Kean Mtn And Grassy Creek / Barr Lake 345 KV; DFAX = 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blue / Black Rock 138 KV Line	Rebuild line with higher-capacity conductor	\$5,623,000	8% (\$438,000)
Black Rock / Nashua 138 KV	Rebuild line with higher-capacity conductor	\$11,969,000	8% (\$933,000)
Batesville / Affinity 138 KV line	Rebuild line with higher-capacity conductor	\$17,493,000	6% (\$1,125,000)
Parke / Taloga 138 KV	Rebuild line with higher-capacity conductor	\$8,344,000	5% (\$433,000)
Sandy Ridge / Coal Creek 138 KV Line	Rebuild line with higher-capacity conductor	\$4,890,000	6% (\$313,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
American Falls / Falconer 138 KV	Rebuild line with higher-capacity conductor	\$3,661,000	4% (\$152,000)
Sandy Ridge / N Reading 138 KV	Rebuild line with higher-capacity conductor	\$3,531,000	6% (\$226,000)
Reata / American Falls Substation 138 KV line	Rebuild line with higher-capacity conductor	\$5,432,000	4% (\$200,000)
Reata Substation / Glenoveer 138 KV	Rebuild line with higher-capacity conductor	\$6,911,000	4% (\$255,000)
N Reading / loco 138 KV	Rebuild line with higher-capacity conductor	\$2,087,000	6% (\$133,000)
Glenoveer / Greens Bayou 69 138 KV	Rebuild line with higher-capacity conductor	\$4,729,000	4% (\$174,000)
Kean Mtn And Grassy Creek 138-1 kV Auto	Replace transformer with higher-rated unit	\$3,903,000	6% (\$219,000)
Kean Mtn And Grassy Creek 345/1 kV TX	Replace transformer with higher-rated unit	\$17,832,000	6% (\$1,002,000)
Dayton 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$16,182,000	7% (\$1,075,000)
Winnebago Wind Power Project / Clinch Riv 138 KV line	Rebuild line with higher-capacity conductor	\$6,452,000	4% (\$293,000)
Poplar Gap / Sandy Hook 138 KV	Rebuild line with higher-capacity conductor	\$4,929,000	6% (\$317,000)
Weser Substation / Itmann 138 KV line	Rebuild line with higher-capacity conductor	\$4,034,000	6% (\$260,000)
Baileysville / Itmann 138 KV	Rebuild line with higher-capacity conductor	\$16,828,000	6% (\$1,083,000)
No Troy / Lost Nation 138 KV	Rebuild line with higher-capacity conductor	\$5,264,000	3% (\$147,000)
Wesmer / Sandy Hook 138 KV L	Rebuild line with higher-capacity conductor	\$3,350,000	6% (\$216,000)
loco Substation / Kean Mtn And Grassy Crk Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,330,000	6% (\$213,000)
Coal Creek / Claypool Hill 138 KV	Rebuild line with higher-capacity conductor	\$5,971,000	9% (\$555,000)
Lynn / Lost Nation 138 KV	Rebuild line with higher-capacity conductor	\$6,420,000	3% (\$179,000)
Lebanon / Winnebago Wind Power Project Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,460,000	4% (\$157,000)
Rogue / Lynn 138 KV	Rebuild line with higher-capacity conductor	\$14,520,000	3% (\$404,000)
	Total allocated to project		\$10,502,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.